CHAPTER 1

INTRODUCTION

1.1 Objective

The main objective of this project is to develop an IoT-Based Smart Public Restroom Monitoring System that ensures effective hygiene management through real-time monitoring of restroom conditions. The system aims to detect key parameters such as usage frequency, odor levels, water overflow, and environmental conditions to maintain cleanliness and prevent unhygienic situations. It also focuses on providing timely alerts to cleaning staff for quick maintenance and improved user experience. The project further aims to enable remote monitoring through cloud integration, allowing authorities to access real-time data via web or mobile applications. Additionally, it seeks to optimize cleaning schedules, reduce manual inspection efforts, and improve resource utilization. The overall goal is to support smart and sustainable sanitation management, ensuring better public health, hygiene standards, and efficient maintenance of public restroom facilities.

1.2 Purpose of the Project

Traditional public restroom maintenance systems rely on manual inspection and fixed cleaning schedules, which often fail to reflect real-time conditions. This may lead to poor hygiene, unpleasant odor, water overflow, and delayed cleaning. The purpose of this project is to develop an intelligent monitoring system that:

- Tracks restroom usage continuously
- Monitors odor levels and environmental conditions
- Detects water overflow or leakage
- Sends automatic alerts to maintenance staff
- Provides remote access through web or mobile applications

Through this system, the interaction between users and restroom management becomes more efficient and automated, reducing manual effort and improving hygiene standards. The system enhances sanitation by continuously monitoring environmental parameters such as temperature and humidity, which can affect cleanliness and user comfort. It also stores historical data in the cloud, allowing authorities to analyze usage patterns, identify high-traffic periods, and optimize cleaning schedules. This system ensures improved hygiene management, reduced manual effort, faster response to maintenance issues, and enhanced user experience in public restroom facilities.

1.3 Scope of the Project

The scope of this project is to develop an intelligent IoT-Based Smart Public Restroom Monitoring System that enhances hygiene management through real-time monitoring and automated alert mechanisms. The system integrates various sensors such as gas sensors, IR sensors, water level sensors, and temperature-humidity sensors with a microcontroller (ESP32) to continuously monitor restroom conditions. By combining sensor data with cloud connectivity, the system provides accurate detection of odor levels, usage frequency, and overflow conditions to maintain cleanliness and prevent unhygienic situations.

This system can be implemented in public places such as transportation hubs, shopping malls, hospitals, educational institutions, and commercial buildings, where maintaining restroom hygiene is critical. It promotes better sanitation practices by enabling timely cleaning actions and reducing dependency on manual inspection. The system also helps in optimizing maintenance schedules and improving overall facility management efficiency. In the future, the system can be enhanced with advanced features such as AI-based hygiene prediction, automated cleaning mechanisms, user feedback integration, and smart ventilation control systems. Additional sensors like ammonia sensors or occupancy tracking modules can be integrated to further improve monitoring accuracy. Machine learning algorithms can also be incorporated to predict peak usage times and optimize resource allocation.

The project also includes the development of a user-friendly web or mobile interface that allows authorities to monitor restroom conditions remotely. The dashboard displays real-time data, alerts, and historical trends in an easy-to-understand format, ensuring accessibility even for non-technical users. Furthermore, the scope covers secure cloud communication, ensuring that data transmitted from sensors to the cloud is protected through proper encryption and authentication mechanisms. This ensures reliability and data privacy, making the system suitable for both public and commercial use. Another important aspect of the scope is scalability and flexibility.

The system is designed to allow easy integration of additional sensors and features without major modifications, making it adaptable to future technological advancements. Finally, the project includes performance testing and validation under different environmental conditions to ensure accuracy, reliability, and efficient operation. The system is also designed to be cost-effective and energy-efficient, ensuring practical implementation and affordability for large-scale deployment.

1.4 Need for the System

In the current public infrastructure environment, restroom maintenance is primarily carried out through manual inspection and fixed cleaning schedules. While these methods are commonly used, they do not reflect the actual real-time condition of restrooms. In high-traffic areas such as transportation hubs, shopping malls, hospitals, and educational institutions, restrooms are used frequently and can quickly become unhygienic. This often leads to unpleasant odor, water overflow, poor sanitation, and discomfort for users. Hence, there is a significant need for an innovative IoT-based system that automates restroom monitoring and improves hygiene management efficiency.

One of the major challenges in public sanitation is the inability to monitor restroom conditions continuously in real time. Parameters such as odor levels, humidity, water leakage, and usage frequency are not tracked effectively through manual methods. Without a proper alert mechanism, cleaning staff may not be informed about the actual condition of the restroom, resulting in delayed maintenance. A smart restroom monitoring system with real-time tracking and automated alerts ensures that maintenance staff are immediately notified about unhygienic conditions, thereby improving cleanliness and user safety.

Additionally, poor sanitation in public restrooms can lead to the spread of diseases and health risks. Unclean environments can promote bacterial growth and create unsafe conditions for users. An intelligent system that continuously monitors environmental conditions and alerts authorities can help maintain proper hygiene standards and prevent such risks.

Another important need for the system is to reduce manual effort and improve operational efficiency. Regular inspection of multiple restrooms requires significant time and manpower. An automated IoT-based system reduces the dependency on manual supervision by identifying only the areas that require immediate attention, thereby optimizing resource utilization and improving maintenance efficiency.

Furthermore, with the growing development of smart cities, there is an increasing demand for integrating IoT technology into public infrastructure. A smart restroom monitoring system supports this transformation by enabling remote monitoring, cloud-based data access, and data-driven decision-making. This enhances overall facility management and ensures better service quality.

Finally, public facilities and commercial establishments require reliable systems to maintain hygiene standards and improve user experience. An IoT-based monitoring system ensures continuous tracking, faster response time, reduced human error, and efficient sanitation management, making it essential for modern public infrastructure.

1.5 Significance of the Project

The IoT-Based Smart Public Restroom Monitoring System introduces an intelligent and automated approach to sanitation management in public facilities. Its significance lies in transforming conventional restroom maintenance methods into smart monitoring systems capable of tracking hygiene conditions such as usage, odor levels, water overflow, and environmental parameters in real time. By integrating IoT sensors, cloud connectivity, automated alerts, and data analysis, the project enhances cleanliness, improves maintenance efficiency, and ensures better public health standards in both public and commercial environments.

One of the most important aspects of this project is its ability to improve hygiene and reduce health risks. In many public places, restrooms are not cleaned on time due to lack of proper monitoring, leading to the spread of bacteria and infections. By providing real-time alerts about unhygienic conditions such as bad odor, overflow, or excessive usage, the system ensures timely cleaning and prevents potential health hazards. This contributes to safer and more hygienic public environments.

The project also plays a significant role in improving resource management and reducing manual effort. Traditional maintenance requires frequent manual inspection, which is time-consuming and inefficient. The proposed system automates monitoring and notifies only when necessary, allowing cleaning staff to focus on areas that require attention. This leads to better utilization of manpower, reduced operational costs, and increased efficiency.

From a technological perspective, the system enhances convenience through remote monitoring and cloud-based access. Authorities can monitor restroom conditions in real time using web or mobile dashboards, enabling faster decision-making and improved facility management. Public infrastructures such as malls, hospitals, railway stations, and educational institutions can maintain higher hygiene standards more effectively.

Furthermore, the system supports the development of smart cities by integrating IoT technology into public sanitation systems. The use of real-time data, automated alerts, and analytics helps in improving overall infrastructure management. In future, the system can be expanded with features such as AI-based prediction, automated cleaning systems, smart ventilation control, and user feedback integration, making it a scalable and future-ready solution.

Overall, the project ensures improved sanitation, enhanced user experience, efficient maintenance, and better public health safety, making it highly significant in modern urban environments.

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CHAPTER 2

LITERATURE SURVEY

2.1 IoT-Based Public Hygiene Monitoring System

Publication Year: 2019

Authors: M. Reddy, P. Sharma, et al.

Technology/Focus: Gas Sensors, Humidity Sensors, Arduino

Summary:

This research focuses on monitoring hygiene levels in public places using IoT-enabled sensors. The system detects odor, humidity, and temperature variations to determine cleanliness levels. Alerts are generated when conditions become unhygienic. The study highlights that continuous monitoring helps prevent poor sanitation but lacks advanced analytics for predicting future conditions.

2.2 Smart Sanitation System Using IoT and Cloud Computing

Publication Year: 2020

Authors: R. Kumar & S. Patel

Technology/Focus: IoT, Cloud Storage, Real-Time Alerts

Summary:

This paper proposes a smart sanitation system that monitors environmental conditions in public restrooms and transmits data to a cloud platform. The system provides real-time alerts to maintenance staff through mobile applications. The study emphasizes improved response time and better hygiene maintenance. It also discusses the importance of cloud-based data storage for remote monitoring and analysis.

2.3 IoT-Based Smart Restroom Monitoring System

Publication Year: 2021

Authors: A. Verma, K. Singh, et al.

Technology/Focus: IoT Sensors, ESP8266, Cloud Monitoring

Summary:

This research presents an IoT-based system for monitoring restroom conditions using sensors such as gas sensors, water level sensors, and motion sensors. The system collects real-time data and sends alerts when parameters exceed predefined thresholds. The study highlights that automated monitoring improves cleaning efficiency and reduces manual inspection efforts. However, the system mainly focuses on threshold-based detection rather than predictive analysis.

2.4 Cloud-Based Smart Facility Monitoring System

Publication Year: 2021

Authors: L. Fernandez & T. Joseph

Technology/Focus: Cloud Computing, Remote Monitoring, IoT

Summary:

This research presents a cloud-based monitoring system for managing public facilities, including restrooms. The system enables remote access to real-time data and supports decision-making through dashboards and alerts. The study highlights improved operational efficiency and reduced manual effort. It also emphasizes the role of secure cloud communication in ensuring data reliability.

2.5 Smart Toilet System with Automated Maintenance Alerts

Publication Year: 2022

Authors: J. Lee & H. Kim

Technology/Focus: Embedded Systems, IoT, Automation

Summary:

This study introduces a smart toilet system that automatically detects usage and cleanliness levels. The system sends notifications to cleaning staff when maintenance is required. It also includes automation features such as ventilation control and water management. The research demonstrates improved efficiency but focuses mainly on automation rather than data-driven decision-making.

2.6 IoT and Data Analytics for Smart City Sanitation

Publication Year: 2023

Authors: S. Gupta, N. Iyer

Technology/Focus: IoT, Data Analytics, Smart City Infrastructure

Summary:

This paper explores the integration of IoT and data analytics in urban sanitation systems. It highlights how real-time data collection and analysis can improve public hygiene management. The study emphasizes the importance of historical data analysis for identifying usage patterns and optimizing cleaning schedules. It also suggests future enhancements using predictive analytics and AI.

CHAPTER 3

EXISTING AND PROPOSED SYSTEM

3.1 Existing System

In traditional public restroom management systems, maintenance is primarily carried out through manual inspection and fixed cleaning schedules. Cleaning staff are assigned to check restrooms at regular intervals, regardless of the actual usage or current condition. Although this approach is widely used, it often fails to maintain proper hygiene standards, especially in high-traffic areas. As a result, restrooms may remain unclean for extended periods, leading to unpleasant odor, water overflow, and poor sanitation.

Some existing systems have introduced basic monitoring using sensors and microcontrollers. These systems can detect parameters such as gas levels, humidity, or water overflow and may provide alerts when values exceed predefined limits. However, most of these solutions are limited to monitoring individual parameters and do not provide a comprehensive view of restroom conditions. Additionally, many systems lack integration with cloud platforms, making remote monitoring and real-time access difficult.

Furthermore, existing systems often do not consider usage patterns or provide intelligent analysis for efficient maintenance planning. The absence of centralized monitoring and automated alert mechanisms results in delayed response and inefficient resource utilization.

Studies indicate that the lack of real-time monitoring and automation leads to poor hygiene management, increased maintenance effort, and reduced user satisfaction. Therefore, there is a need for an advanced, integrated solution that combines multi-sensor monitoring, real-time alerts, cloud connectivity, and intelligent data analysis to ensure efficient and effective restroom maintenance.

3.2 Limitations of Existing System

The review of existing systems reveals several key limitations:

- Dependence on manual inspection of restroom conditions.
- Lack of continuous real-time monitoring of usage and hygiene levels.
- Limited or delayed alert mechanisms for maintenance staff.

- Absence of integrated cloud-based remote monitoring systems.
- No intelligent analysis of usage patterns or predictive maintenance.
- Higher risk of poor sanitation and health hazards due to delayed cleaning.

Additionally, most existing restroom systems do not provide data logging or historical analysis, making it difficult to track usage patterns or optimize cleaning schedules. These limitations highlight the need for an advanced IoT-based monitoring system.

Existing systems also lack integration with smart devices and remote notification features, which reduces awareness for maintenance staff when they are not physically present. Without automated alerts and real-time monitoring, issues such as odor, overflow, or excessive usage may go unnoticed for long periods.

3.3 Proposed System

The proposed IoT-Based Smart Public Restroom Monitoring System is designed to overcome the limitations of existing systems by integrating real-time monitoring, automated alerts, and cloud-based remote access into a single intelligent platform. The system uses sensors such as gas sensors, IR sensors, water level sensors, and temperature-humidity sensors along with a microcontroller (ESP32) and Wi-Fi connectivity to continuously monitor restroom conditions. The system automatically generates alerts when parameters such as odor levels, water overflow, or usage exceed predefined limits. Notifications are sent to maintenance staff through mobile applications or alert systems, ensuring timely cleaning and improved hygiene. Cloud integration enables secure data storage, remote monitoring, and historical data analysis through dashboards.

Furthermore, the system can be expanded with advanced features such as AI-based hygiene prediction, smart ventilation control, automated cleaning mechanisms, and user feedback integration. This makes the project scalable and future-ready. By combining IoT technology, automation, and intelligent alerts, the proposed system improves sanitation, reduces manual effort, enhances user experience, and supports smart city infrastructure.

Key Features of the Proposed System:

- Real-time monitoring of restroom conditions using IoT sensors
- Detection of odor levels, usage frequency, and water overflow
- Wi-Fi-based cloud connectivity for remote access

- Automated alerts and notifications for maintenance staff
- Secure cloud storage for data logging and analysis
- User-friendly dashboard for monitoring restroom status
- Scalable architecture for future AI and smart city integration
- Intelligent threshold-based alert system
- Low-power and energy-efficient system design
- Integration capability with additional sensors for advanced monitoring

Conclusion for Existing and Proposed System :

From the study of the existing system, it is clear that traditional restroom maintenance methods rely mainly on manual inspection and fixed cleaning schedules, which are inefficient in handling real-time conditions. These methods often result in delayed cleaning, poor sanitation, unpleasant environments, and increased health risks due to the absence of continuous monitoring and automated alert mechanisms.

proposed **IoT-Based Smart Public Restroom Monitoring System** addresses these limitations by integrating multiple sensors, real-time data collection, and cloud-based monitoring into a single intelligent platform. The system continuously tracks key parameters such as usage, odor levels, water overflow, and environmental conditions, and generates instant alerts when maintenance is required. This enables timely response from cleaning staff, reduces manual effort, and ensures consistent hygiene standards.

Furthermore, the inclusion of cloud connectivity allows remote monitoring, data storage, and analysis, which helps in understanding usage patterns and optimizing maintenance schedules. The system is scalable and can be enhanced with advanced features such as AI-based prediction and smart automation in the future.

Overall, the proposed system provides a reliable, efficient, and modern solution for public restroom management, contributing to improved sanitation, better resource utilization, enhanced user satisfaction, and the development of smart and sustainable infrastructure.

CHAPTER 4

HARDWARE & SOFTWARE REQUIRED

4.1 Hardware Required

To develop and implement the IoT-Based Smart Public Restroom Monitoring System efficiently, the following hardware components are required:

Microcontroller: ESP32 (30-pin) – Acts as the main processing unit to collect sensor data, process it, and transmit it to the cloud using built-in Wi-Fi.

IR Sensor: Used to detect user presence and monitor restroom usage frequency. **Temperature & Humidity Sensor:** DHT11 (2 units) – Used to measure environmental conditions such as temperature and humidity inside the restroom.

Gas Sensor: MQ135 – Used to detect harmful gases and odor levels, helping to determine hygiene conditions.

Display Unit: 0.96 OLED Display (SSD1306) – Used to display real-time data such as sensor values and system status.

LED Indicators: Red, Green, Blue LEDs – Used to indicate system status and alert conditions (e.g., normal, warning, critical).

Buzzer: Used to provide audible alerts when abnormal conditions like bad odor or overflow are detected.

Breadboard (Large): Used for circuit connections and prototyping.

Connecting Wires:

- Male to Female wires (10)
- Male to Male wires (10)
- Rainbow wire (1m)
- Lead wires (1m)

These are used for establishing connections between components.

Power Supply: 5V power source to power the ESP32 and connected sensors.

The above hardware components ensure effective monitoring of restroom conditions such as usage, odor levels, and environmental parameters. The system supports real-time alert generation and reliable operation. The components used are cost-effective, easy to integrate, and suitable for both prototype and real-time implementation. Additionally, the modular design allows future expansion with more sensors and advanced features.

4.2 Software Required

The proposed system integrates IoT monitoring, real-time data processing, cloud communication, and mobile notification services, requiring the following software tools and technologies:

Operating System

- Windows 10 / 11 (recommended)
- Linux (Ubuntu) or macOS (optional support)

Programming & Development Tools

- Arduino IDE – For programming ESP32 microcontroller
- Embedded C / C++ – For implementing sensor logic, alert system, and control operations

Libraries Used

- WiFi.h – For internet connectivity
- BlynkSimpleEsp32.h – For cloud communication
- Wire.h – For I2C communication
- Adafruit_GFX.h & Adafruit_SSD1306.h – For OLED display
- DHT.h – For temperature and humidity sensing

Cloud & IoT Platforms

- Blynk IoT Platform – For real-time data monitoring, mobile dashboard, and alerts
- Firebase / AWS IoT (Optional) – For advanced cloud storage and scalability

Frontend Technologies

- HTML5 – Interface structure
- CSS3 / Bootstrap – Responsive design
- JavaScript – Dynamic updates and visualization

Backend Technologies (Optional for Advanced System)

- Node.js / Python (Flask) – Server-side processing
- MySQL / MongoDB – Database for storing sensor data and logs
- REST API – Communication between hardware and application

Notification & Alert Services

- Blynk Event Notification – For sending real-time alerts (temperature, gas, overcrowding
- Push Notification APIs .

These software tools support real-time monitoring, automated alerts, cloud-based data handling, and remote access. The system is scalable and can be enhanced with AI-based prediction and smart city integration in future.

CHAPTER 5

SYSTEM DESIGN

5.1 Block Diagram

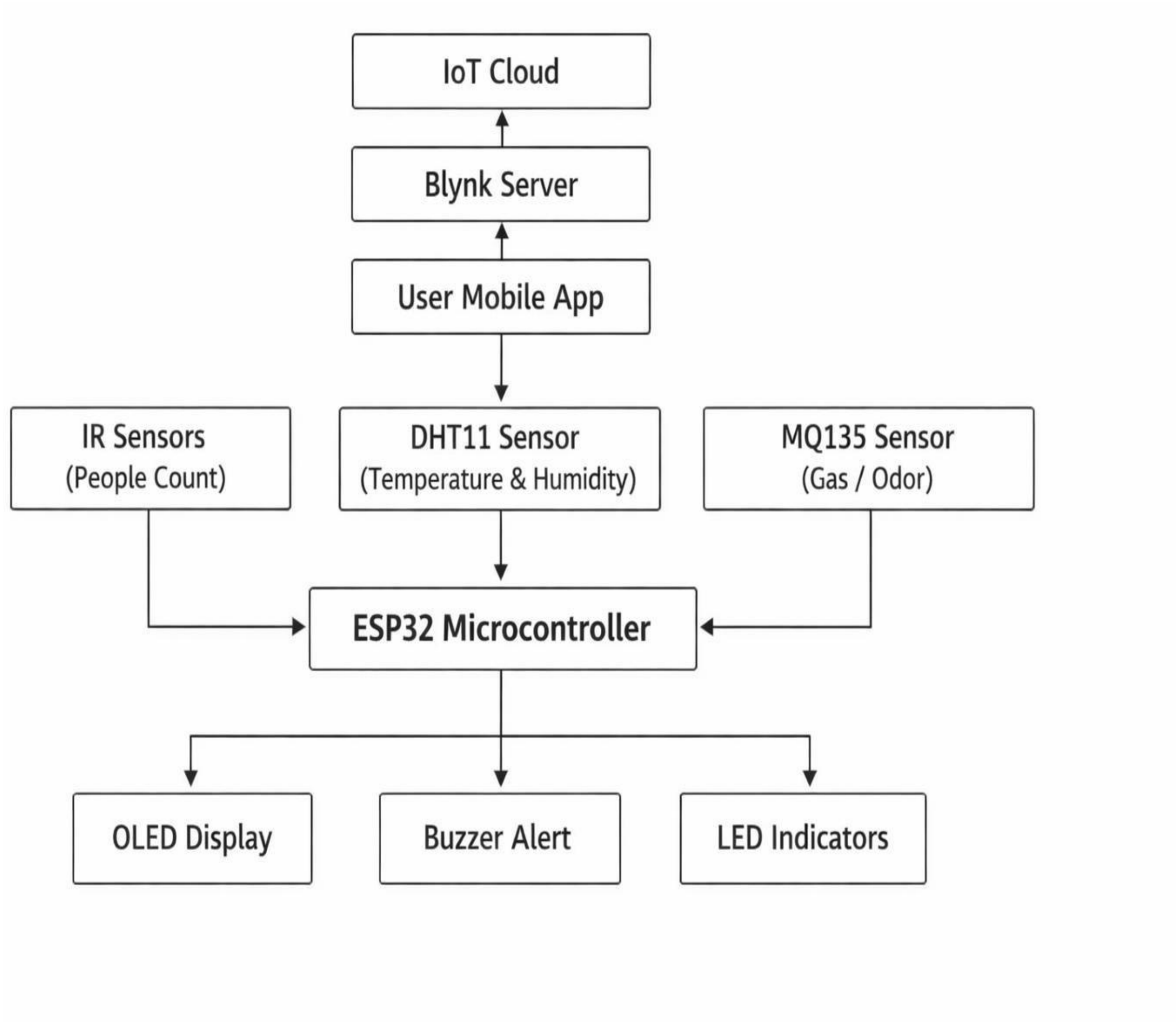

Fig 5.1.1 : IOT-BASED SMART PUBLIC RESTROOM MONITORING SYSTEM

5.1 Explanation of System Architecture

The proposed IoT-Based Smart Public Restroom Monitoring System operates by integrating multiple sensors, a microcontroller, cloud storage, and a user application to monitor restroom conditions in real time. At the top level, the IoT Cloud stores and manages all restroom data through the Blynk Server. The server processes the collected information and makes it accessible to the User App, where users or maintenance staff can monitor temperature, odor levels, and usage status remotely from anywhere. Inside the restroom system, the IR Sensors are used to detect user movement and count the number of people entering the restroom, helping to analyze usage frequency and overcrowding. The DHT11 Sensor continuously measures temperature and humidity levels to ensure a comfortable and hygienic environment. Additionally, the MQ135 Gas Sensor monitors air quality and detects unpleasant odors, which indicate poor sanitation conditions.

All these sensors send their data to the ESP32 Microcontroller, which acts as the central control unit of the system. The microcontroller processes the collected sensor data and determines the overall condition of the restroom. Based on predefined threshold values, the system performs a sanitation status analysis. If abnormal conditions such as high temperature, bad odor, or excessive usage are detected, the system activates the Alerts & Notification module.

The alert system includes a buzzer and LED indicators for local warnings, while notifications are also sent to the mobile application through the Blynk platform. This ensures immediate action by maintenance staff to clean or manage the restroom effectively.

Overall, the system provides continuous monitoring, reduces manual inspection, improves hygiene standards, and ensures a better user experience through real-time IoT-based automation. Additionally, the system enhances operational efficiency by maintaining historical data in the cloud, allowing authorities to analyze usage patterns and optimize cleaning schedules. The modular design enables easy integration of additional sensors such as water level sensors or advanced AI- based prediction systems in the future.

The continuous synchronization between sensors, the microcontroller, and the cloud platform ensures real-time data updates with minimal delay, making the system reliable, scalable, and suitable for smart city infrastructure.

CHAPTER 6

MODULE DESCRIPTION

6.1 Temperature & Humidity Monitoring Module

This module continuously monitors the environmental conditions inside the restroom using a DHT11 sensor. It measures real-time temperature and humidity levels and sends the data to the ESP32 microcontroller at regular intervals. These parameters are important for maintaining hygiene, user comfort, and preventing the growth of bacteria in moist environments. The ESP32 analyzes the sensor data by comparing it with predefined threshold values. If the temperature or humidity exceeds safe limits, the system detects it as an abnormal condition and activates alerts such as a buzzer, LED indicators, and mobile notifications through the IoT platform. This ensures immediate attention to unfavorable conditions. In addition, the module helps identify issues like poor ventilation, excess moisture, or water leakage, which may lead to bad odor and unhygienic surroundings. Continuous data logging to the cloud allows analysis of environmental trends, helping improve cleaning schedules and maintenance efficiency. Overall, this module provides reliable, real-time monitoring and supports automated restroom management by reducing manual inspection and ensuring better sanitation control. Temperature Monitoring Module

6.2 Usage Monitoring Module (People Count)

This module is responsible for monitoring the number of users entering the restroom using infrared (IR) sensors. The sensors detect human movement and count entries based on directional logic, and the data is transmitted to the ESP32 microcontroller for processing.

Tracking usage is essential to understand the load on the restroom facility. As the number of users increases, the probability of the restroom becoming unhygienic also increases. The system continuously updates the usage count and compares it with predefined threshold limits. When the count exceeds the limit, it indicates overcrowding or heavy usage, and the system prepares to trigger alerts for cleaning or maintenance. In addition to real-time monitoring, this module helps in analyzing usage patterns such as peak hours, average usage, and idle periods. The collected data is stored in the cloud, allowing authorities to optimize cleaning schedules based on actual usage rather than fixed time intervals. This reduces unnecessary cleaning efforts and ensures efficient resource utilization. Moreover, the module improves operational efficiency by reducing manual supervision. It enables predictive maintenance by identifying high-usage trends, thereby ensuring that cleaning staff are deployed at the right time. This enhances hygiene levels and improves user satisfaction.

6.3 Gas Detection Module

This module is designed to monitor air quality inside the restroom using the MQ135 gas sensor. It detects harmful gases, ammonia, and unpleasant odors that indicate poor sanitation conditions. The sensor continuously measures gas concentration levels and sends the data to the ESP32 microcontroller. The system compares the gas values with predefined threshold limits. If the gas level exceeds the safe limit, it is identified as an unhygienic condition, and alert mechanisms are activated. This module plays a critical role in maintaining a clean and odor-free environment. Early detection of bad odor helps in taking immediate action before the condition worsens. It improves user comfort and ensures a better restroom experience. Additionally, continuous monitoring allows the system to log gas level data over time. This helps in identifying recurring sanitation problems and areas that require frequent cleaning. The module contributes significantly to improving hygiene standards and maintaining air quality.

6.4 Data Processing Module

The Data Processing Module acts as the core of the system and is controlled by the ESP32 microcontroller. It collects input data from all sensors including the DHT11 (temperature & humidity), IR sensors (usage count), and MQ135 (gas detection). The microcontroller processes this data using programmed logic written in Embedded C/C++ through the Arduino IDE. It performs real-time analysis by comparing sensor values with predefined threshold conditions. Based on this analysis, the system determines the overall restroom status such as Normal, Warning, or Critical Condition. If any parameter exceeds its limit, the system identifies it as a sanitation risk and initiates alert mechanisms. Furthermore, this module ensures accurate decision-making by filtering noise from sensor data and maintaining stable output. It integrates all hardware components into a single automated system and ensures smooth operation. By automating data analysis and decision-making, this module reduces human intervention, improves system reliability, and enhances operational efficiency.

6.5 Alert & Notification Module

This module is responsible for generating alerts when abnormal conditions are detected in the restroom. Alerts are provided through both local and remote mechanisms:

- Local alerts using buzzer and LED indicators
- Remote alerts through mobile application (Blynk platform)

The system generates different types of alerts such as High Temperature, Bad Smell, and Overcrowding based on sensor inputs. These alerts help maintenance staff take immediate action. Additionally, the alert system can differentiate between warning-level and critical-level conditions. Early-stage alerts notify users before the situation becomes severe, enabling preventive maintenance.

The module ensures timely response, reduces health risks, and improves overall sanitation management. It enhances efficiency by ensuring that cleaning is performed only when required.

6.6 IoT Communication Module

The IoT Communication Module enables real-time communication between the system and the cloud using the built-in Wi-Fi capability of the ESP32. Sensor data such as temperature, humidity, gas levels, and usage count are transmitted to the Blynk cloud platform. This allows real-time monitoring and remote access through a mobile application. The module also supports cloud data storage, enabling historical data analysis. Users can review past records to identify trends such as peak usage times and recurring sanitation issues. Secure data transmission ensures reliability and protects user information. This module enhances scalability and makes the system suitable for smart city infrastructure and large-scale deployments.

6.7 User Interface Module

This module provides interaction between the user and the system through an OLED display and mobile application. The OLED display shows real-time parameters such as temperature, humidity, gas levels, usage count, and system status within the restroom. It provides instant visual feedback. The mobile application (Blynk) allows users to monitor data remotely, receive alerts, and check system status in real time. It presents information in a simple and user-friendly format. Additionally, the interface can include graphical representations, status indicators, and notifications to enhance user understanding. It allows quick decision-making and improves system usability. The module ensures smooth communication between the system and users, making the entire system easy to operate, efficient, and highly effective.

CHAPTER 7

CONCLUSION & FUTURE ENHANCEMENT

7.1 Conclusion

The project “IoT-Based Smart Restroom Monitoring System” successfully demonstrates the effective integration of sensor technology, microcontroller-based processing, and cloud connectivity to improve hygiene management in public and private restrooms. By utilizing sensors such as DHT11 for temperature and humidity, MQ135 for gas detection, and IR sensors for user count monitoring, the system continuously observes environmental and usage conditions in real time. The ESP32 microcontroller processes this data efficiently and enables intelligent decision-making based on predefined threshold values.

One of the key achievements of this system is its ability to automate restroom monitoring without the need for constant manual supervision. The system detects unfavorable conditions such as high temperature, excessive humidity, bad odor, or overcrowding and immediately triggers alerts through buzzer indications, LED signals, and mobile notifications via IoT platforms like Blynk. This ensures timely cleaning and maintenance actions, thereby improving hygiene standards and user comfort.

The integration of IoT technology further enhances the system by enabling remote monitoring and real-time data access through mobile applications. Facility managers can view live environmental data, receive instant alerts, and analyze historical records to optimize cleaning schedules and resource utilization. This not only reduces human effort but also increases operational efficiency and reliability.

Additionally, the system promotes preventive maintenance by identifying issues such as poor ventilation, excessive usage, and sanitation degradation at an early stage. The modular design allows easy scalability, making it suitable for deployment in various environments such as schools, hospitals, malls, and public restrooms.

Overall, this project provides a smart, cost-effective, and scalable solution for modern sanitation challenges. It enhances cleanliness, reduces health risks, and supports smart city initiatives by leveraging IoT technology. The system also lays a strong foundation for future advancements such as AI-based cleanliness prediction, automated cleaning mechanisms, and integration with larger smart infrastructure systems, making it a forward-looking and impactful solution

7.2 Future Enhancement

In the future, the Smart Refrigerator Food Expiry Alert System can be enhanced by integrating advanced sensors such as gas sensors (e.g., MQ series) to detect spoilage based on odor or chemical emissions from food items. This would improve the accuracy of the system by identifying actual spoilage conditions rather than relying only on time and temperature. Additionally, incorporating humidity sensors can further refine environmental monitoring, ensuring optimal storage conditions for different types of food.

Artificial Intelligence (AI) and Machine Learning (ML) techniques can also be integrated to make the system more intelligent and adaptive. By analyzing user consumption patterns, storage habits, and historical data, the system can predict food expiry more accurately and suggest optimal storage durations. A camera-based food recognition system can be implemented to automatically detect and classify food items placed inside the refrigerator, eliminating manual input and improving user convenience.

The system can be expanded into a fully integrated smart home solution by connecting with voice assistants such as Alexa or Google Assistant. This would allow users to receive alerts, check food status, or control the system through voice commands. Mobile applications can be further improved with advanced features such as graphical dashboards, multilingual support, and personalized recommendations for better user interaction and accessibility.

Furthermore, cloud-based big data analytics can be used to monitor long-term storage patterns and optimize both food management and energy consumption. Integration with smart energy systems can help regulate refrigerator power usage based on load and usage patterns, contributing to energy efficiency and sustainability.

In commercial environments, the system can be scaled to monitor multiple refrigeration units from a centralized dashboard, improving inventory management and ensuring compliance with food safety standards. Future enhancements may also include predictive maintenance features that analyze system performance and alert users about potential hardware issues, thereby reducing maintenance costs and increasing appliance lifespan.

Overall, these advancements will transform the system into a fully automated, intelligent, and eco-friendly solution, supporting smart living, reducing food wastage, and enhancing overall efficiency in both household and industrial applications.

APPENDIX A
(SOURCE CODE)

```
#define BLYNK_TEMPLATE_ID "TMPL3XYZU_KPQ"
#define BLYNK_TEMPLATE_NAME "Smart restroom monitoring system"

Y_CQSz8WsUtHX0Uk1KcrG5jDCHlefdwS

#define BLYNK_TEMPLATE_ID "TMPL3XYZU_KPQ"
#define BLYNK_TEMPLATE_NAME "Smart toilet monitoring system"
#define BLYNK_AUTH_TOKEN "Y_CQSz8WsUtHX0Uk1KcrG5jDCHlefdwS"

#include <WiFi.h>
#include <BlynkSimpleEsp32.h> #include <Wire.h>
#include <Adafruit_GFX.h> #include <Adafruit_SSD1306.h> #include <DHT.h>

// WiFi
char ssid[] = "pooja@123"; char pass[] = "12345678";

// OLED
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);

// IR
#define IR_A 14
#define IR_B 33

// DHT
#define DHTPIN 4
#define DHTTYPE DHT11
```

```
DHT dht(DHTPIN, DHTTYPE);

// MQ135
#define MQ135_PIN 34

// Alert
#define BUZZER 13
#define RED_LED 12

int count = 0;
bool A_detected = false; bool lockState = false;

unsigned long lastTempRead = 0; float temp = 0;

unsigned long lastBlink = 0; bool blinkState = false;

bool alertSent = false; BlynkTimer timer;

//  FAST IR COUNT (A → B ONLY)
void handlePeopleCount() { bool A = digitalRead(IR_A); bool B = digitalRead(IR_B);

if (lockState) {
if (A == HIGH && B == HIGH) {
lockState = false; A_detected = false;
}
return;
```

```
}

if (A == LOW && !A_detected) { A_detected = true;
}

if (B == LOW && !A_detected) { lockState = true;
}

if (A_detected && B == LOW) { count++;
lockState = true;
}
}

// NON-BLOCKING TEMP
void readTemp() {
if (millis() - lastTempRead > 1000) { float t = dht.readTemperature();
if (!isnan(t)) temp = t - 1.5; lastTempRead = millis();
}
}

// MAIN FUNCTION (SLOW TASKS)
void sendData() {

readTemp();
int gasValue = analogRead(MQ135_PIN);

bool alert = false;
String reason = "NORMAL";
```

```
if (temp > 33) { alert = true;
  reason = "HIGH TEMP";
  }
else if (gasValue > 800) { alert = true;
  reason = "BAD SMELL";
  }
else if (count > 10) { alert = true;
  reason = "OVER CROWD";
  }

  //  BLINK ALERT
  if (alert) {
if (millis() - lastBlink > 300) { blinkState = !blinkState; digitalWrite(BUZZER, blinkState);
  digitalWrite(RED_LED, blinkState); lastBlink = millis();
  }
  } else {
  digitalWrite(BUZZER, LOW); digitalWrite(RED_LED, LOW);
  }

  //  BLYNK
  Blynk.virtualWrite(V0, temp); Blynk.virtualWrite(V1, gasValue); Blynk.virtualWrite(V2, count);
  Blynk.virtualWrite(V3, reason); Blynk.virtualWrite(V4, alert);

  if (alert && !alertSent) {


  Blynk.logEvent("toilet_alert", reason); alertSent = true;
  }
  if (!alert) alertSent = false;
```

```
// OLED UI
display.clearDisplay(); display.setTextSize(1);

display.setCursor(0, 0);  display.print("Temp: "); display.print(temp);

display.setCursor(0, 10);
display.print("Gas: "); display.print(gasValue);

display.setCursor(0, 20); display.print("Count: "); display.print(count);

display.setCursor(0, 35);  display.print("Status: "); display.print(reason);

if (alert) { display.setCursor(0, 50);
display.print("Need to Clean!");
}

display.display();
}

// ⚙️ SETUP void setup() {
Serial.begin(115200);

pinMode(IR_A, INPUT_PULLUP); pinMode(IR_B, INPUT_PULLUP);

pinMode(BUZZER, OUTPUT); pinMode(RED_LED, OUTPUT);

dht.begin();

display.begin(SSD1306_SWITCHCAPVCC, 0x3C); display.clearDisplay();
display.setTextColor(WHITE);
```

```
// WELCOME SCREEN
display.setTextSize(1); display.setCursor(0, 20); display.print("Welcome to Smart");

display.setCursor(0, 30); display.print("Restroom!");

display.setCursor(0, 45); display.print("Keep it Clean :)");

display.display(); delay(3000);

Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

timer.setInterval(1000L, sendData);
}

// LOOP (FAST TASKS)
void loop() {
handlePeopleCount();   // ultra fast counting

Blynk.run();
timer.run();
}
```

APPENDIX A
(SCREEN SHOTS)

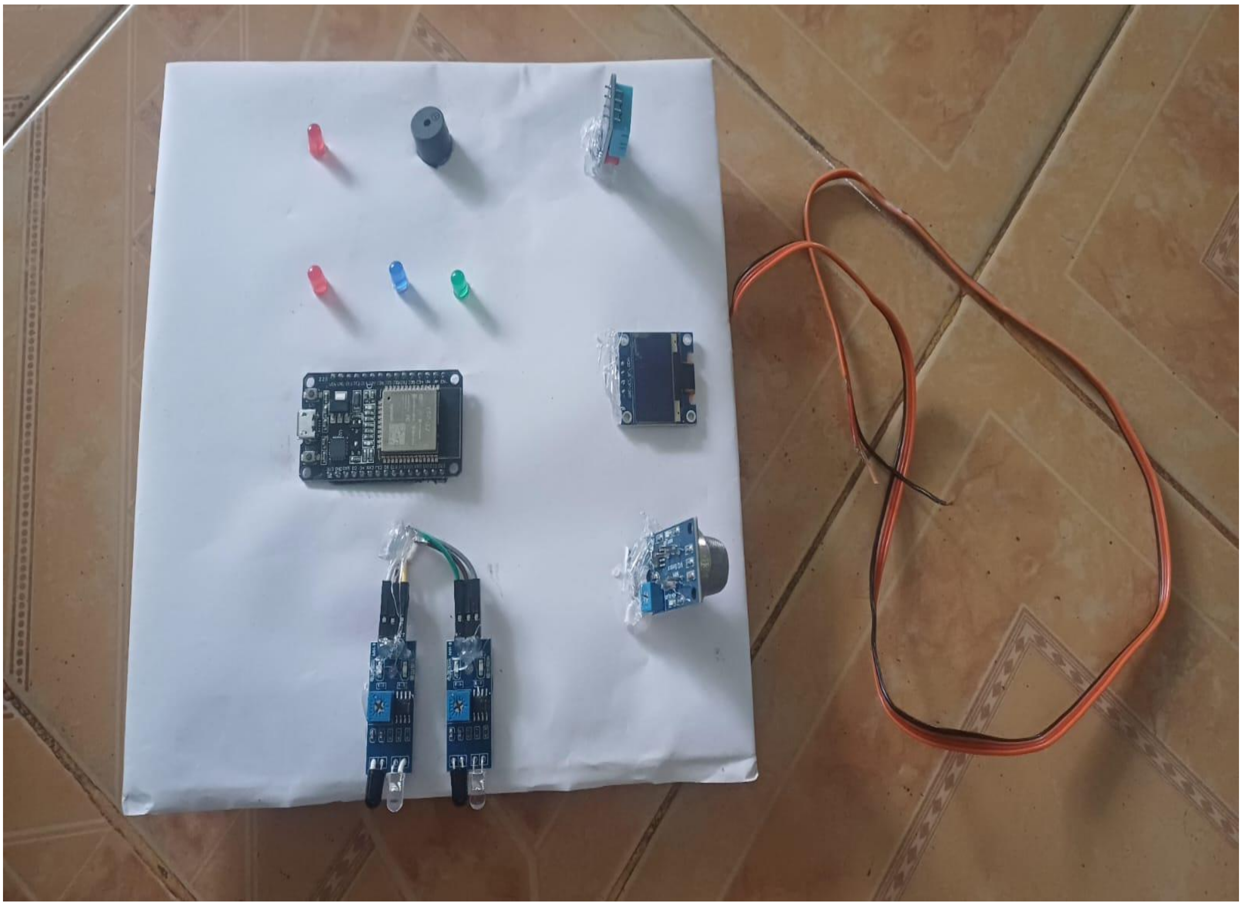

Fig B.1 Smart public restroom monitoring system - TOP VIEW

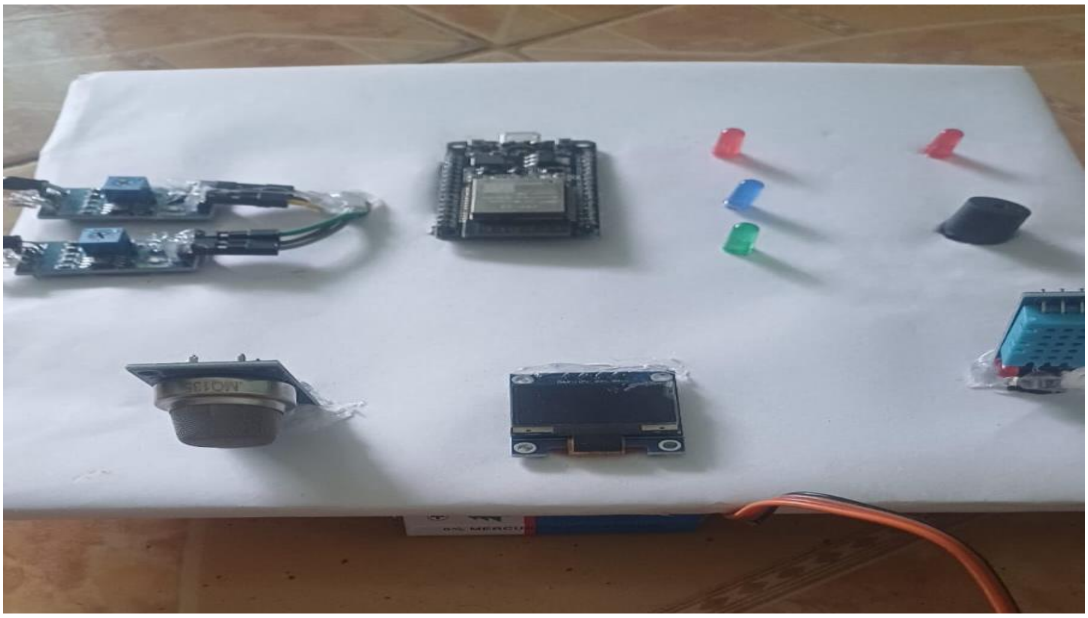

Fig B.2 Smart public restroom monitoring system- FRONT VIEW

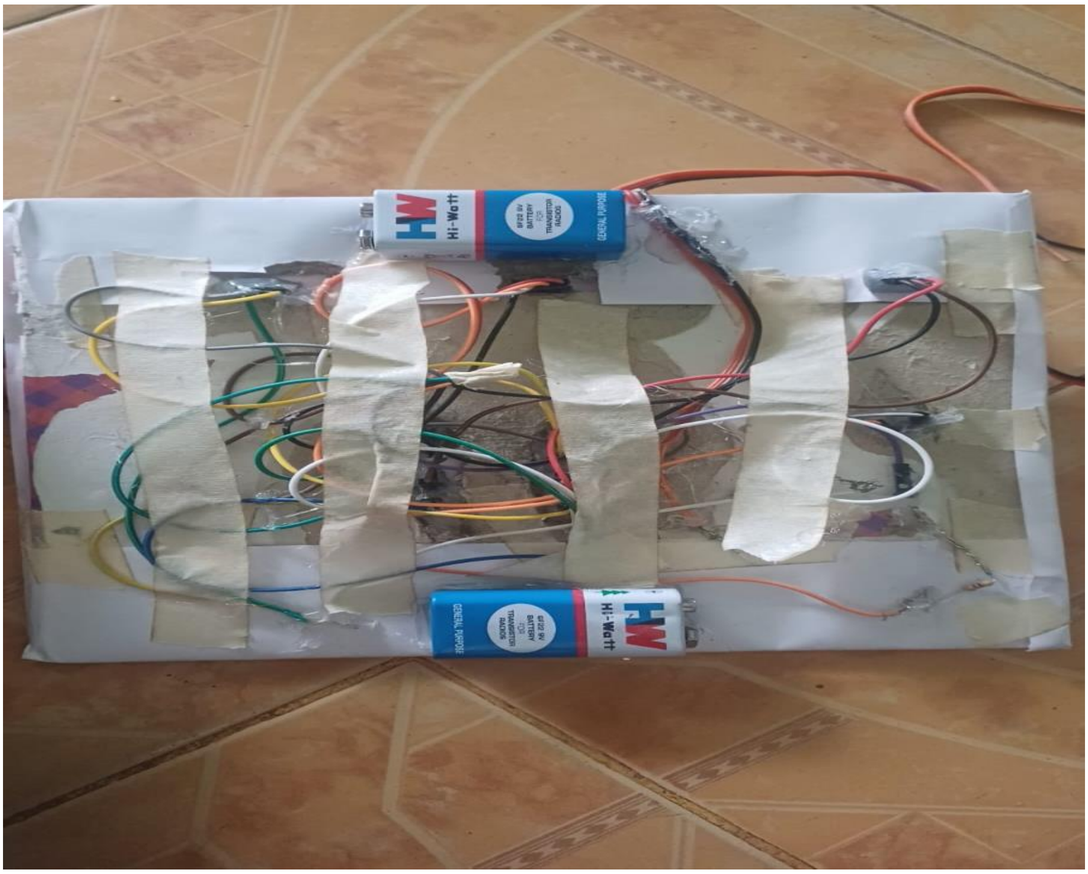

Fig B.3 Smart public restroom monitoring system- BACK VIEW

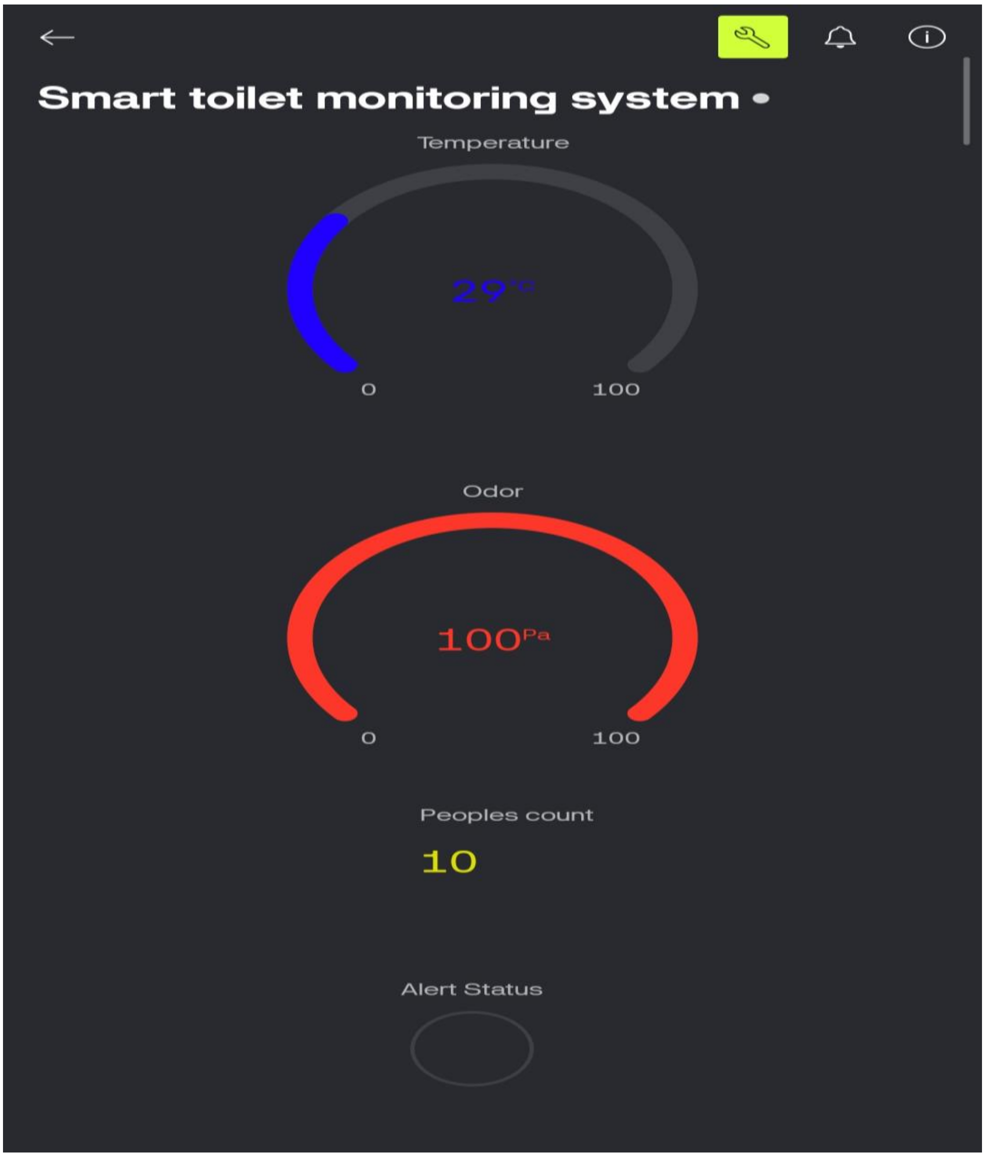

Fig B.4 Smart public restroom monitoring system- DASHBOARD

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